Standard Normal Distribution Reference Sheet

How to read the table.

Let

$$Z \sim \mathcal{N}(0,1) \qquad \text{with} \qquad \Phi(z) = P(Z \le z)$$

The table gives the area to the left of a positive z-value under the standard normal curve. The left column gives the ones and tenths digits of z, while the top row gives the hundredths digit.

Example: For

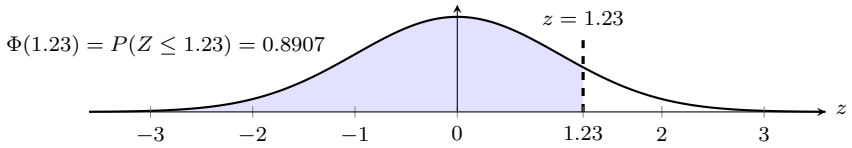
$$\Phi(1.23) = P(Z \le 1.23) = 0.8907$$

look for row 1.2 and column 0.03.

Useful identities

$$P(Z > z) = 1 - \Phi(z),$$
$$\Phi(-z) = 1 - \Phi(z),$$
$$P(a < Z < b) = \Phi(b) - \Phi(a),$$
$$P(|Z| > a) = 2[1 - \Phi(a)].$$

Φ : Phi $\qquad$ ϕ : phi



Probability density function

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Standard normal table: values of $\Phi(z) = P(Z \le z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

Student's t-Distribution Reference Sheet

How to read the table.

Let

$$T \sim t_\nu, \qquad \text{where } \nu \text{ is the number of degrees of freedom (df)}$$

For a sample of size n, we have that $\nu = n - 1$. The table gives positive critical values $t^*_{\alpha,\nu}$, defined by

$$P\big(T > t^*_{\alpha,\nu}\big) = \alpha$$

The left column gives the degrees of freedom ν. Select the column corresponding to the desired tail probability α.

Useful identities

$$P\big(T > t^*_{\alpha,\nu}\big) = \alpha,$$
$$P\big(T < -t^*_{\alpha,\nu}\big) = \alpha,$$
$$P\big(|T| > t^*_{\alpha,\nu}\big) = 2\alpha,$$
$$P\big(-t^*_{\alpha,\nu} < T < t^*_{\alpha,\nu}\big) = 1 - 2\alpha.$$

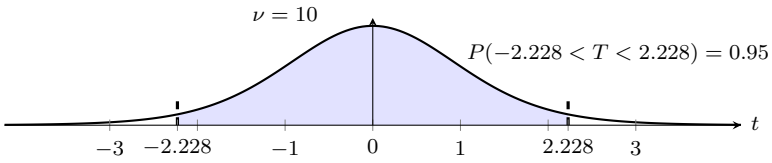
α : alpha $\qquad$ ν : nu

Example: For $\nu = 10$ and $\alpha = 0.025$,

$$t^* = 2.228$$

so $\qquad P(T > 2.228) = P(T < -2.228) = 0.025$

and therefore $\qquad P(-2.228 < T < 2.228) = 1 - 2(0.025) = 0.95.$



Probability density function

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Student's t critical values t^*

ν	Tail probability α						
df	0.30	0.20	0.10	0.05	0.025	0.01	0.005
1	0.727	1.376	3.078	6.314	12.706	31.821	63.657
2	0.617	1.061	1.886	2.920	4.303	6.965	9.925
3	0.584	0.978	1.638	2.353	3.182	4.541	5.841
4	0.569	0.941	1.533	2.132	2.776	3.747	4.604
5	0.559	0.920	1.476	2.015	2.571	3.365	4.032
6	0.553	0.906	1.440	1.943	2.447	3.143	3.707
7	0.549	0.896	1.415	1.895	2.365	2.998	3.499
8	0.546	0.889	1.397	1.860	2.306	2.896	3.355
9	0.543	0.883	1.383	1.833	2.262	2.821	3.250
10	0.542	0.879	1.372	1.812	2.228	2.764	3.169
11	0.540	0.876	1.363	1.796	2.201	2.718	3.106
12	0.539	0.873	1.356	1.782	2.179	2.681	3.055
13	0.538	0.870	1.350	1.771	2.160	2.650	3.012
14	0.537	0.868	1.345	1.761	2.145	2.624	2.977
15	0.536	0.866	1.341	1.753	2.131	2.602	2.947
16	0.535	0.865	1.337	1.746	2.120	2.583	2.921
17	0.534	0.863	1.333	1.740	2.110	2.567	2.898
18	0.534	0.862	1.330	1.734	2.101	2.552	2.878
19	0.533	0.861	1.328	1.729	2.093	2.539	2.861
20	0.533	0.860	1.325	1.725	2.086	2.528	2.845

ν	Tail probability α						
df	0.30	0.20	0.10	0.05	0.025	0.01	0.005
21	0.532	0.859	1.323	1.721	2.080	2.518	2.831
22	0.532	0.858	1.321	1.717	2.074	2.508	2.819
23	0.532	0.858	1.319	1.714	2.069	2.500	2.807
24	0.531	0.857	1.318	1.711	2.064	2.492	2.797
25	0.531	0.856	1.316	1.708	2.060	2.485	2.787
26	0.531	0.856	1.315	1.706	2.056	2.479	2.779
27	0.531	0.855	1.314	1.703	2.052	2.473	2.771
28	0.530	0.855	1.313	1.701	2.048	2.467	2.763
29	0.530	0.854	1.311	1.699	2.045	2.462	2.756
30	0.530	0.854	1.310	1.697	2.042	2.457	2.750
35	0.529	0.852	1.306	1.690	2.030	2.438	2.724
40	0.529	0.851	1.303	1.684	2.021	2.423	2.704
45	0.528	0.850	1.301	1.679	2.014	2.412	2.690
50	0.528	0.849	1.299	1.676	2.009	2.403	2.678
60	0.527	0.848	1.296	1.671	2.000	2.390	2.660
80	0.526	0.846	1.292	1.664	1.990	2.374	2.639
100	0.526	0.845	1.290	1.660	1.984	2.364	2.626
120	0.526	0.845	1.289	1.658	1.980	2.358	2.617
200	0.525	0.843	1.286	1.653	1.972	2.345	2.601
∞	0.524	0.842	1.282	1.645	1.960	2.326	2.576